

Hui Shen

huishen@umich.edu • [Webpage](#) • [Google Scholar](#) • [GitHub](#)

Research Interests

Efficient AI, Agentic AI, Model Evaluation

Education

University of Michigan – Ann Arbor, MI

Ph.D. Student in Computer Science

Advisor: [Z. Morley Mao](#)

Sep 2025

– May 2030 (Exp.)

The Ohio State University – Columbus, OH

B.S. in Computer & Information Science

GPA: 3.95/4.0; Major GPA: 4.0/4.0; Summa Cum Laude

Advisor: [Mi Zhang](#) and [Jingbo Meng](#)

Dec 2024

Work Experience

Research Scientist Intern

Zoom

San Jose, CA, USA

Work with Dr. [Linyong Nan](#), Dr. [Yuwei Fang](#), and Dr. [Alex Acero](#).

May 2026

– Present

Research Assistant II

The University of Hong Kong

Hong Kong, SAR

Work with Prof. [Ngai Wong](#), Prof. [Ping Luo](#), and Prof. [Lingpeng Kong](#).

Jan 2025

– Aug 2025

Software Development Engineer Intern

Swire Coca-Cola China

Hangzhou, China

Jun 2022

– Aug 2022

Publications

Selected Publications

[Dynamic Linear Attention](#) ↗

1

Xin Wang*, **Hui Shen***, Boyuan Zheng, Xueshen Liu, Minkyung Cho, Zhongwei Wan, Zesen Zhao, Zhuoqing Mao, Mi Zhang, Shen Yan.
ICML 2026.

[PhyX: Does Your Model Have the "Wits" for Physical Reasoning?](#) ↗

2

Hui Shen, Taiqiang Wu, Qi Han, Yunta Hsieh, Jizhou Wang, Yuyue Zhang, Yuxin Cheng, Zijian Hao, Yuansheng Ni, Xin Wang, Zhongwei Wan, Kai Zhang, Wendong Xu, Jing Xiong, Ping Luo, Wenhui Chen, Chaofan Tao, Z. Morley Mao, Ngai Wong.
Under review.

[OS-Omni: A Cross-Platform Benchmark for Generalist](#)

3

[Computer-Using Agents](#) ↗

Hui Shen, Yunta Hsieh, Jianing Ma, Ziyuan Liu, Qi Han, Xiuqi Xu, Yanheng Shang, Chenqi Yin, Zesen Zhao, Boyuan Zheng, Jisen Li, Xiaoxia Wu, Ruoyan Zhang, Dongfu Jiang, Ruiyao Liu, Jing Xiong, He Xiao, Nan Zheng, Hangxiang Chen, Shengyang Tao, Deyang Luorong, Jingxuan Zhang, Chaofan Tao, Yuren Wang, Jiaqi Mo, Hei Ting Una Chan, Sicheng Chen, Kai Zhang, Mengkang Hu, Zhongwei Wan, Xin Wang, Chuanyang Zheng, Ziheng Zhang, Hengyuan Zhang, Zunhai Su, Kanzhi Cheng, Jiawei Lu, Yifan Zhang, Shen Yan, Ben Athiwaratkun, Qiushi Sun, Mi Zhang, Ping Luo, Wenhui Chen, Ngai Wong.
Under review.

[SwingArena: Competitive Programming Arena for Long-context](#)

4

[GitHub Issue Solving](#) ↗

Wendong Xu, Jing Xiong, Chenyang Zhao, Qiujiang Chen, Haoran Wang, **Hui Shen**, Zhongwei Wan, Jianbo Dai, Taiqiang Wu, He Xiao, Chaofan Tao, Z. Morley Mao, Ying Sheng, Zhijiang Guo, Hongxia Yang, Bei Yu, Lingpeng Kong, Quanquan Gu, Ngai Wong.
ICLR 2026 (Oral).

Preprints & Under review

**Attention Sink in Transformers: A Survey on Utilization,
Interpretation, and Mitigation** ↗

1

Zunhai Su, Hengyuan Zhang, Wei Wu, Yifan Zhang, Yaxiu Liu, He Xiao, Qingyao Yang, Yuxuan Sun, Rui Yang, Chao Zhang, Keyu Fan, Weihao Ye, Jing Xiong, **Hui Shen**, Chaofan Tao, Taiqiang Wu, Zhongwei Wan, Yulei Qian, Yuchen Xie, Ngai Wong.

Preprint.

**MathGen: Revealing the Illusion of Mathematical Competence through
Text-to-Image Generation** ↗

2

Ruiyao Liu*, **Hui Shen***, Ping Zhang*, Yunta Hsieh, Yifan Zhang, Jing Xu, Sicheng Chen, Junchen Li, Jiawei Lu, Jianing Ma, Jiaqi Mo, Qi Han, Zhen Zhang, Zhongwei Wan, Jing Xiong, Xin Wang, Ziyuan Liu, Hangrui Cao, Ngai Wong.

Preprint.

**MMSpec: Benchmarking Speculative Decoding for Vision-Language
Models** ↗

3

Hui Shen*, Xin Wang*, Ping Zhang*, Yunta Hsieh, Qi Han, Zhongwei Wan, Ziheng Zhang, Jingxuan Zhang, Jing Xiong, Ziyuan Liu, Yifan Zhang, Hangrui Cao, Chenyang Zhao, Mi Zhang.

Preprint.

Speculative Decoding for Multimodal Models: A Survey ↗

4

Yifan Zhang, Yuren Wang, Yunta Hsieh, Xin Wang, Ping Zhang, Ziyi Yang, Jianing Ma, Zesen Zhao, Boyuan Zheng, Hei Ting Una Chan, Jiarui Li, Xueshen Liu, Kunxiao Gao, Ruiyao Liu, Jingxuan Zhang, Junchen Li, Zhongwei Wan, Ziheng Zhang, Jing Xiong, Shatong Zhu, Hangrui Cao, **Hui Shen**.

Preprint.

**DSDR: Dual-Scale Diversity Regularization for Exploration in LLM
Reasoning** ↗

5

Zhongwei Wan, Yun Shen, Zhihao Dou, Donghao Zhou, Yu Zhang, Xin Wang, **Hui Shen**, Jing Xiong, Chaofan Tao, Zixuan Zhong, Peizhou Huang, Mi Zhang.

Preprint.

DoPE: Denoising Rotary Position Embedding ↗

6

Jing Xiong, Liyang Fan, **Hui Shen**, Zunhai Su, Min Yang, Lingpeng Kong, Ngai Wong.

Under review.

LongEmotion: Measuring Emotional Intelligence of Large Language Models in Long-Context Interaction ↗

7

Weichu Liu, Jing Xiong, Yuxuan Hu, Zixuan Li, Minghuan Tan, Ningning Mao, Chenyang Zhao, Zhongwei Wan, Chaofan Tao, Wendong Xu, **Hui Shen**, Chengming Li, Lingpeng Kong, Ngai Wong.

Under review.

Conferences

Locate, Steer, and Improve: A Practical Survey of Actionable Mechanistic Interpretability in Large Language Models ↗

1

Hengyuan Zhang, Zhihao Zhang, Mingyang Wang, Zunhai Su, Yiwei Wang, Qianli Wang, Shuzhou Yuan, Ercong Nie, Xufeng Duan, Qibo Xue, Zeping Yu, Chenming Shang, Xiao Liang, Jing Xiong, **Hui Shen**, Chaofan Tao, Zhengwu Liu, Senjie Jin, Zhiheng Xi, Dongdong Zhang, Sophia Ananiadou, Tao Gui, Ruobing Xie, Hayden Kwok-Hay So, Hinrich Schütze, Xuanjing Huang, Qi Zhang, Ngai Wong.

ACL 2026 Findings.

ATTS: Asynchronous Test-Time Scaling via Conformal Prediction ↗

2

Jing Xiong, Qiujiang Chen, Fanghua Ye, Zhongwei Wan, Chuanyang Zheng, Chenyang Zhao, **Hui Shen**, Alexander Hanbo Li, Chaofan Tao, Haochen Tan, Haoli Bai, Lifeng Shang, Lingpeng Kong, Ngai Wong.

ICLR 2026.

SRPO: Enhancing Multimodal LLM Reasoning via Reflection-Aware Reinforcement Learning ↗

3

Zhongwei Wan, Zhihao Dou, Che Liu, Yu Zhang, Dongfei Cui, Qinjian Zhao, **Hui Shen**, Jing Xiong, Yi Xin, Yifan Jiang, Chaofan Tao, Yangfan He, Mi Zhang, Shen Yan.

NeurIPS 2025.

MEDA: Dynamic KV Cache Allocation for Efficient Multimodal

4

Long-Context Inference ↗

Zhongwei Wan, **Hui Shen**, Xin Wang, Mi Zhang.

NAACL 2025.

MEIT: Multi-Modal Electrocardiogram Instruction Tuning on Large

5

Language Models for Report Generation ↗

Zhongwei Wan, Che Liu, Xin Wang, Chaofan Tao, **Hui Shen**, Zhenwu Peng, Jie Fu, Rossella Arcucci, Huaxiu Yao, Mi Zhang.

ACL 2025 Findings.

Argus: Benchmarking and Enhancing Vision-Language Models for 3D

6

Radiology Report Generation ↗

Che Liu, Zhongwei Wan, Yuqi Wang, **Hui Shen**, Haozhe Wang, Kangyu Zheng, Mi Zhang, Rossella Arcucci.

ACL 2025 Findings.

SVD-LLM V2: Optimizing Singular Value Truncation for Large

7

Language Model Compression ↗

Xin Wang, Samiul Alam, Zhongwei Wan, **Hui Shen**, Mi Zhang.

NAACL 2025.

Famba-V: Fast Vision Mamba with Cross-Layer Token Fusion ↗

8

Hui Shen, Zhongwei Wan, Xin Wang, Mi Zhang.

ECCV 2024 Workshop on Computational Aspects of Deep Learning (Best Paper Award).

Journals

Efficient Diffusion Models: A Survey ↗

1

Hui Shen^{*}, Jingxuan Zhang^{*}, Boning Xiong^{*}, Rui Hu^{*}, Shoufa Chen, Zhongwei Wan, Xin Wang, Yu Zhang, Zixuan Gong, Guangyin Bao, Chaofan Tao, Yongfeng Huang, Ye Yuan, Mi Zhang.

Transactions on Machine Learning Research (TMLR 2025)

Artificial Intelligence of Things: A Survey ↗

2

Shakhrul Iman Siam, Hyunho Ahn, Li Liu, Samiul Alam, **Hui Shen**, Zhichao Cao, Bhaskar Krishnamachari, Mani Srivastava, Mi Zhang.

ACM Transactions on Sensor Networks.

FedAIoT: A Federated Learning Benchmark for Artificial Intelligence of Things ↗

3

Samiul Alam, Tuo Zhang, Tiantian Feng, **Hui Shen**, Zhichao Cao, Dong Zhao, JeongGil Ko, Kiran Somasundaram, Shrikanth S. Narayanan, Salman Avestimehr, Mi Zhang.

Journal of Data-centric Machine Learning Research (DMLR 2025)

Autoregressive Models in Vision: A Survey ↗

4

Jing Xiong, Gongye Liu, Lun Huang, Chengyue Wu, Taiqiang Wu, Yao Mu, Yuan Yao, **Hui Shen**, Zhongwei Wan, Jinfa Huang, Chaofan Tao, Shen Yan, Huaxiu Yao, Lingpeng Kong, Hongxia Yang, Mi Zhang, Guillermo Sapiro, Jiebo Luo, Ping Luo, Ngai Wong.

Transactions on Machine Learning Research (TMLR 2025)

Teaching Experience

Teaching Assistant

Aug 2024

Systems II: Introduction to Operating Systems (CSE 2431), The Ohio State University.

– Dec 2024

Teaching Assistant

Aug 2024

Introduction to Digital Logic (ECE 2060), The Ohio State University.

– Dec 2024

Open Source Projects

Harbor github.com/laude-institute/harbor

Contributor (2000+ stars) — Agent Evaluation System

Implemented the DS-1000 adapter in Harbor, enabling standardized evaluation workflows for DS-1000 within the system.

Honors & Scholarships

CSE Fellowship, University of Michigan.

2025

Best Paper Award in ECCV 2024 Workshop, NVIDIA.

2024

Microsoft for Startups Founders Hub Member, Microsoft.

2024

Services

Conference Reviewers

EMNLP 2025

CVPR 2025, 2026

ICLR 2025, 2026

ICML 2026

ECCV 2026

Journal Reviewers

IEEE Transactions on Mobile Computing (TMC)

IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)